

# Professor Guidance — Passenger Price Elasticity

Date of discussion: 2 October 2026

## Purpose of this document

This document records the methodological guidance received from the professor during the discussion. It is intended to be used as the working reference before making further changes to the elasticity implementation.

## 1. Trips with distance greater than 100 miles

The professor did not instruct us to automatically remove trips above 100 miles. He asked us to examine the relationship between distance and price for such observations. If a substantially larger distance is accompanied by a correspondingly higher price, the observation may be valid and should not be excluded solely because the distance exceeds 100 miles.

## 2. Very large elasticity values

For observations producing very large elasticity values (for example, values on the order of 1,000 or more), the professor asked us to inspect the individual underlying observations rather than immediately applying a threshold or deleting them.

| Field to inspect | Required value            |
|------------------|---------------------------|
| Price (P)        | Actual trip price         |
| Distance (L)     | Actual trip distance      |
| P_base           | Global base price         |
| L_base           | Global base distance      |
| $\Delta P$       | $(P - P\_base) / P\_base$ |
| $\Delta L$       | $(L - L\_base) / L\_base$ |
| $\epsilon$       | $\Delta L / \Delta P$     |

The professor specifically asked for the actual price and distance corresponding to the unusually large elasticity values so that the cause can be understood before deciding on any data-quality rule.

## 3. Three-month price and distance arrays

The professor suggested that, if practical, we provide the arrays of prices and distances from the three-month dataset. The purpose is to allow direct inspection of the underlying observations. Origin–destination pairing is not the focus for this elasticity calculation.

## 4. Hourly time slots

The analysis should use 24 hourly slots: 00:00–01:00, 01:00–02:00, ..., 23:00–00:00. Thus, the 24-hour representation is Hour 0 through Hour 23.

## 5. Unequal sample sizes across weather × hour groups

The number of observations does not need to be the same for every weather condition and hour. Some peak-hour/weather combinations may have many observations while others may have relatively few. This is acceptable. The relevant statistic to calculate is the standard deviation of the elasticity values within each group.

## 6. Normalization

The professor stated that no additional normalization is required.

## 7. Global base price

The professor clarified that the base price should be independent of the time slot and weather condition. Use one flat global  $P_{\text{base}}$  calculated across all relevant slots and weather conditions, rather than a separate base price for each slot.

## 8. Immediate implementation decision

Do not yet implement an automatic >100-mile exclusion or a  $|\Delta P| \geq 0.001$  threshold based only on the preliminary diagnostics. First inspect the actual extreme observations requested by the professor. Any subsequent data-quality or outlier rule should be based on that inspection and further professor guidance.

## Source

Professor discussion transcript, record-1790931196806. The statements above are a structured summary of that discussion and do not add a new methodological rule where the professor did not explicitly provide one.